AQA 3.1.9 Rate equations

AQA A-level Chemistry · Revision questions · Sheet 1 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- rate = $k[A]^m[B]^n$ — orders come from experimental data, not from the balanced equation
- Doubling a concentration: rate × 1 (zero order), × 2 (first order), × 4 (second order)
- Units of k: rearrange k = rate ÷ (concentration terms), with rate in mol dm^{-3} s^{-1}, and cancel
- Rate-determining step = slowest step; species in the rate equation appear in (or before) it
- ln k = $-E_a/RT$ + ln A, so the gradient of ln k against 1/T = $-E_a/R$

Section A: Short-answer questions

01 Solutions of two compounds, W and X, react together in the presence of a soluble catalyst, Y, as shown in the equation $2W + X \rightarrow Z$

When the concentrations of W, X and Y are all doubled, the rate of reaction increases by a factor of four. Which is a possible rate equation for this reaction?

[1 mark]

A rate = $k[W]^2[X]$

B rate = $k[W]^2[Y]$

C rate = $k[X][Y]$

D rate = $k[X][Z]$

02 The rate equation for a reaction is rate = k[P]. Deduce the units of the rate constant, k.

[1 mark]

03 A suggested mechanism for $2NO_2Cl \rightarrow 2NO_2 + Cl_2$ is

Step 1 $NO_2Cl \rightarrow NO_2 + Cl$ (slow)

Step 2 $NO_2Cl + Cl \rightarrow NO_2 + Cl_2$ (fast)

Deduce the rate equation.

[1 mark]

04 In the first 20 s of a reaction, the concentration of a reactant falls from 0.080 mol dm^{-3} to 0.068 mol dm^{-3}. Calculate the mean rate of reaction over this time.

[1 mark]

05 The rate equation for $2NO + O_2 \rightarrow 2NO_2$ is rate = $k[NO]^2[O_2]$. When [NO] = 0.010 mol dm^{-3} and $[O_2]$ = 0.050 mol dm^{-3}, the rate is 7.1×10^{-3} mol dm^{-3} s^{-1}. Calculate k and give its units.

[2 marks]

06 The rate equation for a reaction is rate = $k[A][B]^2$. Deduce the effect on the rate when the concentration of A is doubled and the concentration of B is halved.

[1 mark]

Section B: Longer questions

07 For a reaction, Table 1 shows the value of the rate constant at different temperatures.

Experiment	Temperature / K	Rate constant / s^{-1}
1	$T_1 = 303$	$k_1 = 1.55 \times 10^{-5}$
2	$T_2 = 333$	$k_2 = 1.70 \times 10^{-4}$

This equation can be used to calculate the activation energy, E_a: $\ln(k_1 \div k_2) = (E_a \div R) \times (1 \div T_2 - 1 \div T_1)$

Calculate the value, in kJ mol^{-1}, of the activation energy, E_a. The gas constant, $R = 8.31$ J K^{-1} mol^{-1}

[5 marks]

08 The rate equation for a reaction is rate = $k[L][M]^2$

08.1 Deduce the overall effect on the rate of reaction when the concentrations of both L and M are halved.

[1 mark]

08.2 The rate of reaction is 0.0250 mol dm^{-3} s^{-1} when the concentration of L is 0.0155 mol dm^{-3}. Calculate the concentration of M if the rate constant is 21.3 mol^{-2} dm^6 s^{-1}

[3 marks]

08.3 Define the term *overall order of reaction*.

[1 mark]

09 The thermal decomposition of but-3-en-1-ol is investigated at different temperatures (T).

$CH_2{=}CHCH_2CH_2OH \rightarrow CH_2{=}CHCH_3 + HCHO$

The results are used to calculate the rate constant, k, at each temperature. Table 1 shows some of the results.

T / K	1/T / K^{-1}	k / s^{-1}	ln k
553	1.81×10^{-3}	4.6×10^{-4}	−7.68
563	1.78×10^{-3}	8.4×10^{-4}	−7.08
573		15.6×10^{-4}	
583	1.72×10^{-3}	28.0×10^{-4}	−5.88
593	1.69×10^{-3}	49.9×10^{-4}	−5.30

09.1 Complete Table 1 with the missing values at 573 K.

[1 mark]

09.2 The overall order of the reaction can be deduced from a piece of information in one of the column headings in Table 1. Identify this piece of information and deduce the overall order.

[2 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**C**	*Source: 7405/3 Jun 2018 Q12*	1
02	s^{-1}	*Source: Practice question (written for this set)*	1
03	rate = k[NO_2Cl]	*Source: Practice question (written for this set)*	1
04	6.0×10^{-4} mol dm^{-3} s^{-1}	*Source: Practice question (written for this set)*	1
05	$k = 7.1 \times 10^{-3} \div (0.010^2 \times 0.050) = 1420$ (1.42×10^3) mol^{-2} dm^6 s^{-1}	*Source: Practice question (written for this set)*	2
06	Rate is halved	*Source: Practice question (written for this set)*	1
07	M1 $\ln(1.55 \times 10^{-5} \div 1.70 \times 10^{-4}) = E_a/R \times (1/333 - 1/303)$ M2 $-2.39 = E_a/R \times (-2.97 \times 10^{-4})$ M3 $E_a = 2.39 \times 8.31 \div 2.97 \times 10^{-4}$ M4 = 66 937 (J mol^{-1}) M5 = 66.9 kJ mol^{-1}	*Source: 7405/2 Jun 2022 Q01.7* If only k_1 and k_2 reversed, gives a negative E_a: lose M1 and M5 Allow range 66.3 – 67.1	5
08.1	Falls by a factor of 8 / multiplied by 1/8 / divided by 8	*Source: 7405/2 Nov 2020 Q01.4–01.6* Allow halved then quartered / decreases by 2^3	1
08.2	M1 $[M]^2$ = rate ÷ k[L] M2 $[M]^2 = 0.0250 \div (21.3 \times 0.0155)$ $(= 7.57 \times 10^{-2})$ M3 [M] = 0.275 mol dm^{-3} (min 2 sf)	Using 0.25 instead of 0.025 gives 0.870 — scores 2 Upside-down expression gives 3.63 — scores 2	3
08.3	The sum of the powers / indices to which the concentrations are raised (in the rate equation)	Allow all the orders added	1
09.1	1.75×10^{-3} **and** −6.46	*Source: 7405/3 Jun 2024 Q03.1–03.2* Allow 0.00175 Not other sig figs (e.g. 1.7×10^{-3}, −6.5)	1
09.2	M1 unit of k (is s^{-1}) M2 (order) 1 / first	Mark independently; not just 'k'	2